

A community in formation

Corpus construction and initial analysis of Slovene posts on Bluesky

01

Building the corpus

Why Bluesky and why this corpus

- Bluesky makes public posts available through the AT Protocol (ATProto), with no dedicated research API and no data-access agreement.
- Slovene already has strong reference corpora and the JANES social-media corpus, but Bluesky is not yet covered.
- A small language community is easier to cover widely: broad coverage, not complete coverage. It also needs more care.

Building the corpus in four steps

PHASE 1



Tagged discovery

relay crawl and
live firehose

PHASE 2



Seed authors

every DID seen in
a Slovene-tagged post

PHASE 3



Backfill

DID-resolved, per-PDS
post histories

PHASE 4



Language filter

tag, langid, langdetect
groups validated by hand

379,482 posts were retrieved from 742 discovered authors. Language was then decided post by post, combining the s1 tag with langid and langdetect. The tag alone was not enough; the resulting groups were checked by hand on samples.

The corpus in numbers

MEASURE	VALUE	NOTE
Posts	141,013	deduplicated by AT-URI
Unique authors	432	of 742 discovered
Date range	Aug 2023 to Apr 2026	to 16 April 2026
Median post length	86 characters	replies shorter, at 77
Authors for half the posts	22	see section 02
Core-layer validation	300 of 300	no false positives in this sample

02

What the corpus shows

Before and after the increase

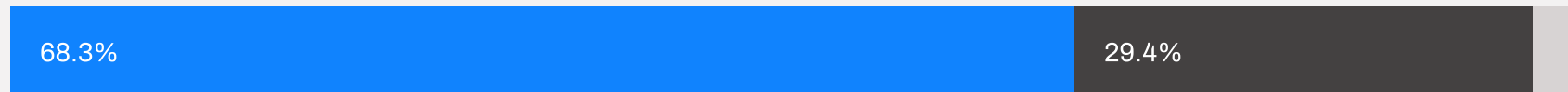
Observed activity in this corpus
October to November 2024

31 → 5,984

Active authors: 13 → 196. Of the 172
authors first observed in November,
86.6% posted again in December and
94.2% within three months.

MEASURE	THROUGH OCT 2024	NOV 2024 ONWARD
Total observed posts	178	140,835
Total active authors	30	429
Median monthly posts	10	7,958
Median monthly active authors	3	210
Median posts per active author	2.33	37.73

Replies, not broadcasts



Replies

Original posts

Quote posts · 2.3%

54.5% of posts were already replies before November 2024

Among the 246 authors with at least 20 posts, **the median reply rate is 0.66** and **only 9 authors never reply**.

This pattern is not caused by just a few accounts.

A small, concentrated core

22 authors

account for half of all 141,013 posts in the corpus.

43 authors

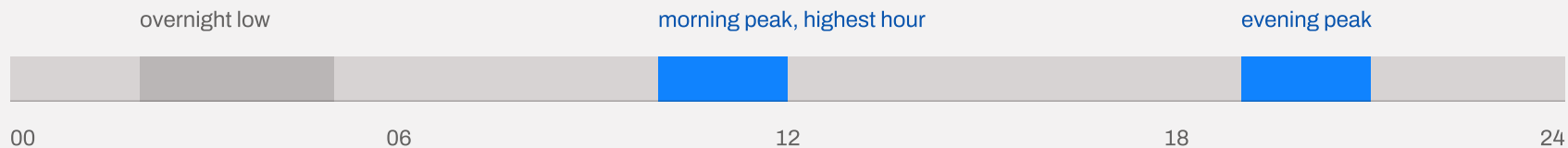
the top tenth, account for 69.6% of the posts.

35 posts

is what the median author contributed, over the whole period.

Uneven participation is common in social-media corpora. It matters more than usual here, because the whole author population is 432 people.

A Slovenian daily rhythm



hours in Europe/Ljubljana time (CET/CEST)

The strip marks the activity windows reported in the paper, not measured hourly counts: two peaks, one trough, consistent with a Central European population. It is a pattern in the corpus, not evidence of where individual authors are.

03

Sharing the corpus

Collecting and sharing are different things

Collecting

Public repository data can be retrieved through Bluesky's protocol without a research access agreement.

That made this corpus possible without special access.

Sharing

When posts are collected together, they become a record of the same people over time.

Handles and custom domains can make authors identifiable, so sharing the corpus needs more care.

How can the corpus be shared?

- The code and collection method are public. The full text is not.
- Sharing only post identifiers would avoid distributing copies of the posts themselves, but it could still recreate a detailed record of this small group.
- The best way to share the corpus is still undecided.

Next steps: live monitoring of the corpus and extending collection and analysis to authors hosted on other personal data servers, such as `eurosky.social` and `w.social`.

Three things to take away

- 01 **METHOD:** Tags helped find candidate accounts; language was then checked post by post.
 - 02 **ACTIVITY:** The late-2024 increase lasted; replies dominate; a small group produces much of the content.
 - 03 **SHARING:** Open access makes collection easier, but sharing a corpus of a small community is more complicated.
-

Thank you



04

Backup

Pipeline detail

Discovery

Historical crawl over the relay's `listReposByCollection`, resolving each repository's DID to its home PDS and fetching records via `listRecords`.

Live firehose subscriber on `subscribeRepos`, handling post creates, updates and deletes plus account-level events. Both stores keyed by AT-URI with soft-deletion tracking.

Backfill

`did:plc` resolved through `plc.directory`, `did:web` through `.well-known/did.json`; the resolved PDS URL cached, then paginated `listRecords` per author.

Not restricted to `bsky.social`: 9 of 432 authors are hosted on `eurosky.social`. Posts already deleted at crawl time are not recoverable and are out of scope.

Decision groups

DECISION GROUP	TAG	LANGID	LANGDETECT	STATUS	POSTS
Core tag-supported	yes	either	either	Included	85,094
Model consensus only	no	yes	yes	Included	46,134
langid only	no	yes	no	Included	5,632
langdetect only	no	no	yes	Included	4,153
Tag only	yes	no	no	Excluded	n/a
Short tagged	yes	not evaluated		Excluded	n/a

Manual annotation of the core layer

300 / 300

All 300 randomly sampled core posts were judged acceptable as Slovene. In the separate review-layer sample, every row from the three included model-supported groups was also acceptable; the 72% overall rate reflects the two groups that were excluded.

False negatives

MEASURE	VALUE
Sampled excluded, untagged posts from seed authors	200
Judged Slovene-dominant or mixed	17 (8.5%)
Comparable excluded pool	97,968
Extrapolated missed Slovene posts	≈ 8,300
Relative to corpus size	≈ 5.9%

Detector comparison

Same observed precision, different recovery

fastText and Lingua were evaluated on the same manually validated samples as langid and langdetect. On the comparable subset all four showed the same observed precision.

The langid/langdetect pair recovered more acceptable Slovene posts, so the original pair was retained for the final build.

Where detector disagreement concentrates

Closely related South Slavic languages: langid labelled 4,506 posts Croatian and 2,285 Bosnian, against only 278 English.

Meanwhile 41.4% of posts carry both sl and en tags: mixed metadata that looks like client-side tagging behaviour rather than English-dominant text.